

Akarsh Jain

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PROFILE

Backend-focused Computer Science undergraduate at SVNIT Surat who builds systems all the way to production – REST APIs, authentication, and background job pipelines backed by automated tests and real CI/CD deployment. Comfortable across the stack from FastAPI/PostgreSQL to Docker and cloud infrastructure (AWS EC2), with judgment sharpened by 600+ problems solved on LeetCode and competitive ratings on Codeforces and CodeChef. Seeking a Software Engineering / Backend Development internship.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Bachelor of Technology (B.Tech), Computer Science and Engineering

2024 – 2028
CGPA: 7.55 / 10

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL
Backend & APIs: FastAPI, REST API Design, JWT/OAuth2, LangGraph, Pydantic, RBAC
Testing & Tools: Pytest, Unit & Integration Testing, Git, GitHub Actions, Docker, Postman
Deployment & Infra: AWS EC2, Render, Caddy, Docker Compose, CORS, Env Configuration
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

SentinelReview – Agentic Security Code Review for GitHub PRs

GitHub

- Python, FastAPI, LangGraph, PostgreSQL/pgvector, Redis/RQ, React, TypeScript, Bandit, Semgrep
- An AI reviewer bot that scans GitHub Pull Requests for security vulnerabilities, explains each finding against authoritative CWE/OWASP sources, and proposes a verified patch – automating the first pass of a security code review.
- Built a 7-agent LangGraph pipeline combining Bandit/Semgrep static analysis with TF-IDF/cosine retrieval; GitHub App JWT (RS256) + OAuth2 auth and HMAC webhooks trigger RQ workers that persist findings end-to-end.
- Reached 0.900 precision / 1.000 recall (0.952 F1) on a hand-labeled benchmark after ingesting 150 live GHSA advisories, and cut analyzer latency 5x via batching, backed by 101 backend + 20 frontend tests.

Mini Code Judge

GitHub – Live Demo

- FastAPI, PostgreSQL, Docker, Redis/RQ, JWT, Gemini API
- A competitive-programming judge in the spirit of Codeforces/LeetCode – write C/C++/Java/Python in-browser, get instant feedback on sample cases, then submit for full judging against hidden test cases across problems, contests, and a leaderboard.
- Runs untrusted submissions in resource-capped, network-isolated Docker sandboxes (memory/CPU/process limits) with an automatic OS-level fallback, verdicts via Accepted/WA/TLE/MLE/RE/CE; submissions queue durably through Redis/RQ to horizontally-scalable workers.
- Multi-provider auth (email verification, Google, GitHub OAuth) with account logout; timed contests with live leaderboards and penalty tracking, plus Gemini-powered AI code review after each submission.

IronLog – Gym Progress Analytics Platform

GitHub – ironlog.in

- FastAPI, SQLAlchemy, JWT, Vanilla JS PWA, Chart.js, Docker, AWS EC2, Caddy, GitHub Actions
- A multi-user fitness tracker that turns raw body-weight, lift, and calorie logs into real trend analysis and coaching feedback, instead of just storing numbers on a chart.
- Computes Epley 1RM, Mifflin-St Jeor BMR/TDEE, and a true maintenance-calorie figure back-calculated from each user's own logged intake vs. real weight change, plus regression-based weight-trend and goal-ETA forecasting.
- Deployed on AWS EC2 via Docker Compose behind Caddy (auto SSL) with GitHub Actions CI/CD; installable PWA with a Gemini-powered AI Coach and period-over-period progress comparisons, covered by 35 Pytest tests.

ACHIEVEMENTS

Secured 1st Rank in ACM Summer Challenge, August 2025.

Participated in national-level coding contests including Epiphany 14.0 and CodeCraft 2025.

COMPETITIVE PROGRAMMING

LeetCode: 650+ Problems Solved

Profile

Codeforces: Rating 1600+

Profile

CodeChef: Rating 1670+

Profile

INTERESTS

Competitive Programming, Software Development, Artificial Intelligence, DevOps and Cloud Infrastructure, Fitness